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TINGKAT WILAYAH JAKARTA TIMUR 2  
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**TEST PROJECT  
NETWORKING**

**IT NETWORK SYSTEMS  
ADMINISTRATION**

# Introduction

Network technology knowledge is becoming essential nowadays for people who want to build a successful career in any IT engineering field. This test project contains a lot of challenges from real life experience, primarily IT integration and IT outsourcing. If you are able to complete this project with a high score, you are definitely ready to service the network infrastructure for any multi-branch enterprise.

## Description of project and tasks

This test project is designed using a variety of network technologies that should be familiar from the Cisco certification tracks. Tasks are broken down into following configuration sections:

- Basic configuration
- Switching
- Routing
- Services
- Security
- WAN and VPN

All sections are independent but all together they build very complex network infrastructure. Some tasks are pretty simple and straightforward; others may be tricky. You may see that some technologies are expected to work on top of other technologies. For example, IPv6 routing is expected to run on top of configured VPNs, which are, in turn, expected to run on top of IPv4 routing, which is, in turn, expected to run on top of PPPoE, and so on. It is important to understand that if you are unable to come up with a solution in the middle of such a technology stack it doesn't mean that the rest of your work will not be graded at all. For example, you may not configure IPv4 routing that is required for VPN because of IP reachability but you can use static routes and then continue to work with VPN configuration and everything that runs on top. You won't receive points for IPv4 routing in this case but you will receive points for everything that you made operational on top as long as functional testing is successful.

## Instructions Notice to the Competitor

Your configuration will be marked with scripts, so therefore we need two important basic configurations:

1. no ip domain-lookup
2. exec-timeout 0 0 on console

Both configurations are already pre configured on all switches and routers, so do not change these configurations.

# Instructions to the Competitor

1. Read all tasks in each section before proceeding with any configuration. The completion of any item may require the completion of any previous or later item.
2. Points are awarded for working configurations only. Test the functionality of all the requirements before you submit the test project. Be careful, because as you configure one part, you may break a previous requirement or configuration.
3. No partial points can be granted for any aspect; all requirements need to be fulfilled to receive the points for the aspect. Some requirements depend on other aspect's requirements, either before or after the current aspect.
4. Save your configurations frequently; accidents do and will happen.
5. Please use industrial best practice where possible!

## Basic configuration

1. Configure hostnames for all network devices as you see on the topology.
2. Configure domain name **lks2026.id** for all network devices on the topology.
3. Configure **Skills39** as a privileged mode **password** for all network devices on the topology.
4. Configure IPv4 address for all network devices as you see on the topology.
5. Configure "GMT+7" with the offset of +7 as a timezone for all network devices.

## Switching

1. Configure VTP version 2 on all switches to synchronize VLANs. It should be possible to modify VLAN database only from L3SW-1. Set the domain name to **lks2026.id** and password **Skills39**. VLAN databases of all the other switches should be synchronized from L3SW-1. VLAN database on all switches should contain following VLANs.
  - a. VLAN 10 with name SRV
  - b. VLAN 20 with name CLI
2. Configure all links between switches as trunk port.
  - a. Configure manual pruning so that only created VLANs are allowed forwarding.
3. Configure EtherChannel between switches.
  - a. Use following port-channel numbers:
    - i. 1 - between switches L3SW-1 and L3SW-2
    - ii. 2 - between switches L3SW-1 and L2SW-1
    - iii. 3 - between switches L3SW-2 and L2SW-2
  - b. The bundled link between L3SW-1 and L3SW-2 shall not use dynamic negotiation protocol.
  - c. The bundled link between L3SW-1 and L2SW-1 use a Cisco proprietary protocol for dynamic negotiation.
  - d. The bundled link between L3SW-2 and L2SW-2 use a 802.3ad protocol for dynamic negotiation.
  - e. L3SW-1 and L3SW-2 should initiate negotiation and the other devices should respond but don't initiate.
4. Spanning tree configuration.
  - a. L3SW-1 should be root bridge of VLAN10. If L3SW-1 goes down L3SW-2 should take over as the root bridge.
  - b. L3SW-2 should be root bridge of VLAN20. If L3SW-2 goes down L3SW-1 should take over as the root bridge.

- c. The traffic from TMT-CLI should pass through L3SW-1. Achieve this by manipulating the cost to its lowest possible value.
- d. Configure port which is connected to end device so that it immediately begins forwarding when connected. Make sure no BPDU is accepted on those ports.

## Routing

1. Configure FHRP on L3SW-1 and L3SW-2.
  - a. Use Hot Standby Router Protocol v2 for VLAN 10.
    - i. L3SW-1 should be used as the default gateway, achieve this by setting the priority to its max value.
    - ii. If L3SW-1 goes down, L3SW-2 should immediately take over as the default gateway.
    - iii. Use 104 as the group number.
    - iv. Use 10.0.10.254 as virtual IPv4 address
  - b. Use a Hot Standby Router Protocol v2 for VLAN 20.
    - i. L3SW-2 should be used as the default gateway, achieve this by setting the priority to its max value.
    - ii. If L3SW-2 goes down, L3SW-1 should immediately take over as the default gateway.
    - iii. Use 204 as the group number.
    - iv. Use 10.0.20.254 as virtual IPv4 address
2. Configure default route with ISP as the next hop on both TMT-1 and TMT-2.
3. Configure EIGRP between TMT routers and TMT L3 switches
  - a. use as number 26
  - b. advertise only the intranet network
  - c. make sure network vlan 10 and 20 can reach TMT-1
  - d. turn off network summarization
4. Configure static default route on both L3SW to TMT-1
5. Configure OSPF.
  - a. Use OSPF area 0 and instance number 26.
  - b. Advertise only Public IP on all routers.

## Services

1. Configure DHCP
  - a. Configure DHCP service for vlan 20 on L3SW-1
  - b. The pool should be named "VLAN20"
  - c. TMT-CLI should be able to obtain IP address automatically.
  - d. All DHCP client should use BRT-SRV as DNS Server.
2. Configure NAT
  - a. Create standard acl with id 1 to permit vlan 20 network
  - b. When vlan 20 network communicate with internet, its IP address should be translated to TMT-1 public facing interface.
3. Configure TMT-1 remote monitoring using SNMP
  - a. Configure snmp read-only community **public**
  - b. Configure snmp read-write community **sysadmin\_only**
  - c. You can verify if the snmp community is either working or not by using the MIB browser on TMT-CLI
4. Configure DNS
  - a. Configure dns service for zone **lks2026.id** on BRT-SRV
  - b. Create following records:
    - i. A record for ns.lks2026.id with value 10.30.230.100
    - ii. A record for www.lks2026.id with value 10.0.10.100
    - iii. NS record for lks2026.id with value ns.lks2026.id
5. Configure Website
  - a. Configure website for **www.lks2026.id** on TMT-SRV
  - b. Turn on HTTP and HTTPS service

## Security

1. Configure SSH version 2 for remote access on TMT-1.
  - a. Configure **all vty lines** to use SSH
  - b. Use local authentication.
    - i. Username user1 with maximum privilege level and secret **Skills39**
    - ii. Username user2 with privilege level 5 and secret **Skills39**
  - c. Ensure only vlan 20 network is allowed to access via SSH. (use earlier ACL)
2. Configure port-security on the port which is connected to TMT-CLI using following parameters:
  - a. Maximum MAC address – 2
  - b. In case of policy violation, security message should be displayed on the console, port should be disabled.

# VPN

1. Configure a site to site GRE Tunnel VPN connection between TMT-1 and BRT-1
  - a. Use tunnel0 interface
  - b. In BRT-1, create static route to vlan 10 and vlan 20 network using TMT-1 tunnel address as the next hop
  - c. In TMT-1, create static route to 10.30.230.0/24 using BRT-1 tunnel address as the next hop

# Configure Table

| Site     | Device  | Interface            | Address          |
|----------|---------|----------------------|------------------|
| Internet | ISP     | GigabitEthernet0/0   | 202.134.0.254/24 |
|          |         | GigabitEthernet1/0   | 202.134.1.254/24 |
|          |         | GigabitEthernet2/0   | 202.134.2.254/24 |
|          |         | GigabitEthernet3/0   | 202.134.3.254/24 |
|          |         | Loopback0            | 1.1.1.1          |
| BARAT    | BRT-1   | GigabitEthernet0/0   | 202.134.0.1/24   |
|          |         | GigabitEthernet0/1   | 10.30.230.254/24 |
|          |         | Tunnel0              | 10.255.255.1/30  |
|          | BRT-2   | GigabitEthernet0/0   | 202.134.1.1/24   |
|          | BRT-SRV | Fa0                  | 10.30.230.100/24 |
| TIMUR    | TMT-1   | GigabitEthernet0/0   | 11.11.11.2/30    |
|          |         | GigabitEthernet0/1   | 12.12.12.2/30    |
|          |         | GigabitEthernet0/2   | 202.134.2.1/24   |
|          |         | Tunnel0              | 10.255.255.2/30  |
|          | TMT-2   | GigabitEthernet0/0   | 11.11.11.6/30    |
|          |         | GigabitEthernet0/1   | 12.12.12.6/30    |
|          |         | GigabitEthernet0/2   | 202.134.3.1/24   |
|          | L3SW-1  | GigabitEthernet1/0/6 | 11.11.11.1/30    |
|          |         | GigabitEthernet1/0/7 | 12.12.12.5/30    |
|          |         | Vlan 10              | 10.0.10.253/24   |
|          |         | Vlan 20              | 10.0.20.253/24   |
|          | L3SW-2  | GigabitEthernet1/0/6 | 11.11.11.5/30    |
|          |         | GigabitEthernet1/0/7 | 12.12.12.1/30    |
|          |         | Vlan 10              | 10.0.10.252/24   |
|          |         | Vlan 20              | 10.0.20.252/24   |
|          | L2SW-1  | Vlan 10              | 10.0.10.200/24   |

|  |         |         |                     |
|--|---------|---------|---------------------|
|  | L2SW-2  | Vlan 10 | 10.0.10.201/24      |
|  | TMT-CLI | Fa0     | 10.0.20.x/24 (DHCP) |
|  | TMT-SRV | Fa0     | 10.0.10.100/24      |



# Topology

